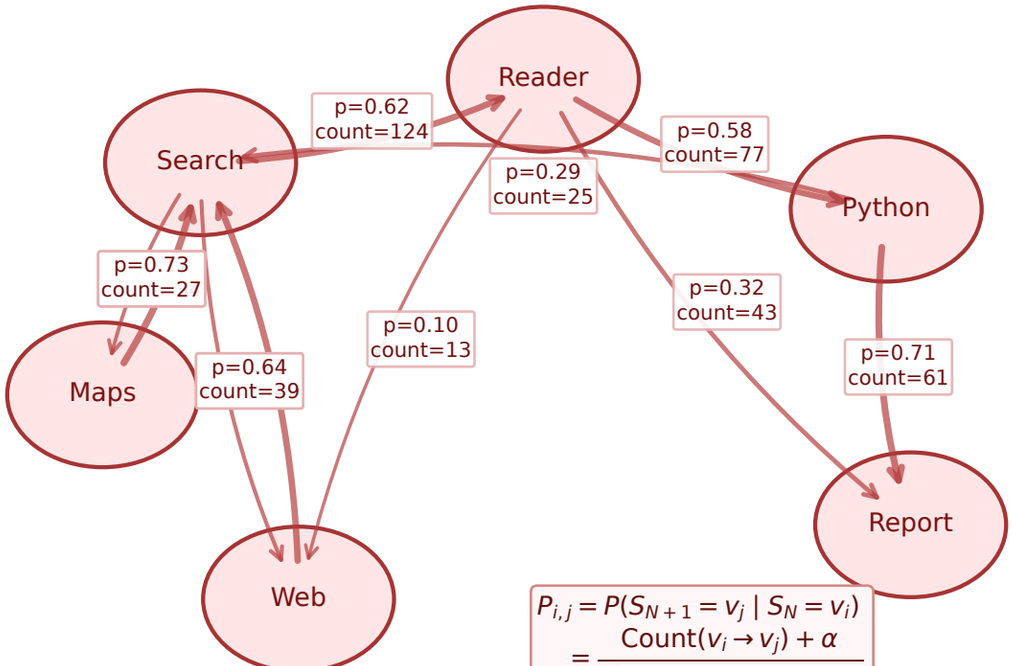
Markov State Transition Graph



$$P_{i,j} = P(S_{N+1} = v_j \mid S_N = v_i) = \frac{\text{Count}(v_i \to v_j) + \alpha}{\sum_k (\text{Count}(v_i \to v_k) + \alpha)}$$

Prefetch Decision (Session Example)

Turn N

Current: Search
Top successors: Reader(0.62), Web(0.21)
Prefetch: Reader

Turn N+1

Observed: Reader
Update SDG edge count Search->Reader
Prefetch: Python

Turn N+2

Observed: Python
Top successors: Report(0.71), Search(0.29)
Prefetch: Report

Turn N+3

Observed: Report
No heavy successor above threshold
No prefetch

Decision policy:
1) Rank successors by transition probability P(i,j)
2) Score by expected gain G(i,j)=P(i,j)*L(j)
3) Keep only P(i,j) >= tau; select top-B